

Eddy Current (EC) Probe for Conductivity Measurements

12-Week Project Calendar

Project goal: Design, optimize, calibrate, and validate an affordable VNA-based eddy-current probe for electrical conductivity measurements.

Week	Main Goal	Key Tasks	Deliverable
1	Background Research	Study eddy currents, conductivity, permeability, skin depth, coil impedance, and VNA S-parameters. Review existing EC probes.	Literature review + project requirements
2	System Specifications	Choose target conductivity and frequency ranges, sample materials, probe dimensions, and desired accuracy.	System specification document
3	Electromagnetic Modeling	Calculate skin depth and select candidate frequencies. Model coil/sample interaction analytically or with simulation software.	Initial frequency + coil design
4	VNA & Measurement Setup	Learn VNA calibration and S-parameter measurement. Develop a measurement procedure and S11-to-impedance conversion.	Working VNA measurement setup
5	Initial Probe Design	Design candidate coil geometries. Consider turns, diameter, wire size, lift-off distance, and operating frequency.	CAD/design files + candidate probes
6	Prototype Fabrication	Build first probe(s). Measure resistance, inductance, Q factor, and frequency response.	Physical prototype
7	Probe Optimization	Test probes on conductive samples. Compare sensitivity to conductivity, lift-off, and permeability.	Optimized probe design
8	Measurement Software	Automate VNA data collection. Extract S11 and calculate impedance or another conductivity-sensitive parameter.	Data collection + analysis workflow
9	Calibration	Measure reference samples with known conductivity. Develop a calibration curve/model.	Calibration model
10	Validation	Measure unknown samples. Compare results with reference values. Calculate error, repeatability, and uncertainty.	Validation results
11	Optimization & Troubleshooting	Investigate lift-off, temperature, probe positioning, permeability, frequency, noise, and VNA drift. Improve the system.	Final optimized system
12	Final Analysis & Presentation	Organize results, create plots, document methods, write conclusions, and prepare presentation/demo.	Final report + presentation

Project Workflow

Research → Specifications → Modeling → VNA Setup → Probe Design → Prototype → Optimization → Data Collection → Calibration → Validation → Final System

Important Experimental Milestones

- Start collecting experimental data as soon as the first prototype works.
- Keep probe geometry, frequency, lift-off distance, and sample temperature documented for every measurement.
- Reserve reference samples for calibration and separate samples for final validation.
- Track repeatability and measurement error throughout the project rather than only at the end.